Effect Systems & Interpreters

Season: 7 · Advanced FP · Category: Functional · Level: Advanced

Intent

- Separate describing effects from running them.

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- Screenplay vs performance.

When to Use

- Serious FP architecture in Java.
- The design force named in Intent shows up more than once.

When Not to Use

- CRUD apps that don't need effect algebras.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.functional.io;

import java.util.function.Function;
import java.util.function.Supplier;

/** Tiny IO effect — suspend side effects, interpret at the edge. */
public final class IO<A> {
    private final Supplier<A> thunk;

    private IO(Supplier<A> thunk) { this.thunk = thunk; }

    public static <A> IO<A> delay(Supplier<A> thunk) { return new IO<>(thunk); }
    public static <A> IO<A> pure(A a) { return new IO<>(() -> a); }

    public <B> IO<B> map(Function<? super A, ? extends B> f) {
        return new IO<>(() -> f.apply(thunk.get()));
    }

    public <B> IO<B> flatMap(Function<? super A, IO<B>> f) {
        return new IO<>(() -> f.apply(thunk.get()).unsafeRun());
    }

    /** Unsafe: runs effects. Call only at the program edge. */
    public A unsafeRun() { return thunk.get(); }
}
```

Modern Java 25 / Functional

- Witness types emulate higher-kinded types.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: functional/io/ + tagless + free
```

Pitfalls

- Encodings are educational — production may use libraries.
- Tagless Final needs Kind encoding without HKTs.

Interview Q&A

- Q: When to pick Effect Systems & Interpreters? A: When you need testable effect boundaries or pure state threading.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Separate describing effects from running them.
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: Screenplay vs performance. [0:40] Intent — State the design force in one sentence: Separate describi...